

Risk and Return:

Expected Returns and Variances and Portfolio Diversification

Chapter 13

Chapter Outline

- Expected Returns and Variances
- Portfolios: Expected Returns and Variances
- Risk: Systematic and Unsystematic
- Diversification and Portfolio Risk
- Total Risk and Standard Deviation
- Systematic Risk and Beta
- Capital Asset Pricing Model (CAPM)

Expected Returns

Individual risky asset

- Expected returns are based on the probabilities of possible outcomes
- In this context, “expected” means average if the process is repeated many times

$$E(R) = \sum P_i R_i$$

Example: Expected return and Standard Deviation

- Consider the following information
 - Invest 50% of your money in Asset A
 - State Probability A B
 - Boom 0.4 30% -5%
 - Bust 0.6 -10% 25%

- What is the expected return and standard deviation for *each asset*?

Example: Expected Return

- What are the expected return for each stock?

- Stock A

- $E(R) = 0.4(0.3) + 0.6(-0.1) = 0.06$

- Stock B

- $E(R) = 0.4(-0.05) + 0.6(0.25) = 0.13$

Variance and Standard Deviation

- Variance and standard deviation still measure the volatility of returns
- Using unequal probabilities for the entire range of possibilities
- Weighted average of squared deviations

$$\sigma^2 = \sum_{i=1}^n p_i (R_i - E(R))^2$$

Example: Variance and Standard Deviation

- What are the variance and standard deviation for each stock?
- Stock A
 - $\sigma^2 = 0.4(0.30-0.06)^2 + 0.6(-0.10-0.06)^2 = ?$
 - $\sigma = ?$
- Stock B
 - $\sigma^2 = 0.4(-0.05-0.13)^2 + 0.6(0.25-0.13)^2 = ?$
 - $\sigma = ?$

Portfolios

- A portfolio is a collection of assets
- An asset's risk and return is important in how it affects the risk and return of the portfolio
- The risk-return trade-off for a portfolio is measured by the portfolio expected return and standard deviation, just as with individual assets

Portfolio Expected Returns (Weighted Approach)

- The expected return of a portfolio is the weighted average of the expected returns for each asset in the portfolio

$$E(R_P) = \sum_{j=1}^m w_j E(R_j)$$

- You can also find the expected return by finding the portfolio return in each possible state and computing the expected value as we did with individual securities

Example: Portfolio Weights

- Suppose you have \$15,000 to invest and you have purchased securities in the following amounts.

What are your portfolio weights in each security?

- | | |
|------------------|----------------------|
| ■ \$2000 of DCLK | •DCLK: $2/15 = .133$ |
| ■ \$3000 of KO | •KO: $3/15 = .2$ |
| ■ \$4000 of INTC | •INTC: $4/15 = .267$ |
| ■ \$6000 of KEI | •KEI: $6/15 = .4$ |

Example: Expected Portfolio Returns (Weighted Approach)

- Consider the portfolio weights computed previously. If the individual stocks have the following expected returns, what is the expected return for the portfolio?
 - DCLK: 19.65%
 - KO: 8.96%
 - INTC: 9.67%
 - KEI: 8.13%
- $E(R_p) = .133(19.65) + .2(8.96) + .267(9.67) + .4(8.13) = 10.24\%$

Portfolio Expected Return and Standard Deviation

- Construct portfolio using the weights
- Compute the expected portfolio return using the same formula as for an individual asset
- Compute the portfolio variance and standard deviation using the same formulas as for an individual asset

Example: Portfolio Expected Return and Standard Deviation

- Consider the following information

- Invest 50% of your money in Asset A

■ State	Probability	A	B	Portfolio
■ Boom	0.4	30%	-5%	12.5%
■ Bust	0.6	-10%	25%	7.5%

- What is the expected return and standard deviation for the *portfolio*?

Risk

- Total risk = systematic (market) risk + (firm specific) unsystematic risk
- The standard deviation of returns is a measure of total risk

Unsystematic (Firm Specific) Risk

- Risk factors that affect a limited number of assets
- Also known as unique risk and asset-specific risk
- Includes such things as labor strikes, part shortages, etc.

Systematic (Market) Risk

- Risk factors that affect a large number of assets
- Also known as non-diversifiable risk or market risk
- Includes such things as changes in GDP, inflation, interest rates, etc.

Measuring Systematic Risk

- How do we measure systematic risk?
- We use the **beta coefficient to measure systematic risk**
- Beta measures a stock's market risk and shows a stock's volatility relative to the market.
- Indicates how risky a stock is if the stock is held in a well-diversified portfolio.

What does beta tell us?

- A beta of 1 implies the asset has the same systematic risk as the overall market
- A beta < 1 implies the asset has less systematic risk than the overall market
- A beta > 1 implies the asset has more systematic risk than the overall market

Beta coefficients of selected companies

Beta Coefficients (β)	
U.S. Companies	
Coca-Cola	0.35
Kellogg	0.57
Microsoft	0.69
Google	1.09
3M	1.24
Tesla	1.28
JC Penney	1.61
Abercrombie & Fitch	1.85
Asian Companies	
Sinopec Shanghai Petroleum	0.48
China Mobile	0.53
PT Telekomunikasi Indonesia	0.84
Taiwan Semiconductor Manufacturing	1.01
HSBC	1.32
Tata Motors	3.17

Diversification

- Portfolio diversification is the investment in several different asset classes or sectors
- **Diversification is not just holding a lot of assets**
- Correlations among assets are important
 - For example, if you own 50 internet stocks, you are not diversified (high correlation)
 - However, if you own 50 stocks that span 20 different industries, then you are diversified (low correlation)

Implications for Portfolio Formation

- Combining assets together with low correlations reduces portfolio risk more
 - The lower the correlation, the lower the portfolio standard deviation
 - Negative correlation reduces portfolio risk greatly
 - Combining two assets with perfect negative correlation reduces the portfolio standard deviation

The Portfolio Standard Deviation

Standard Deviation of Return

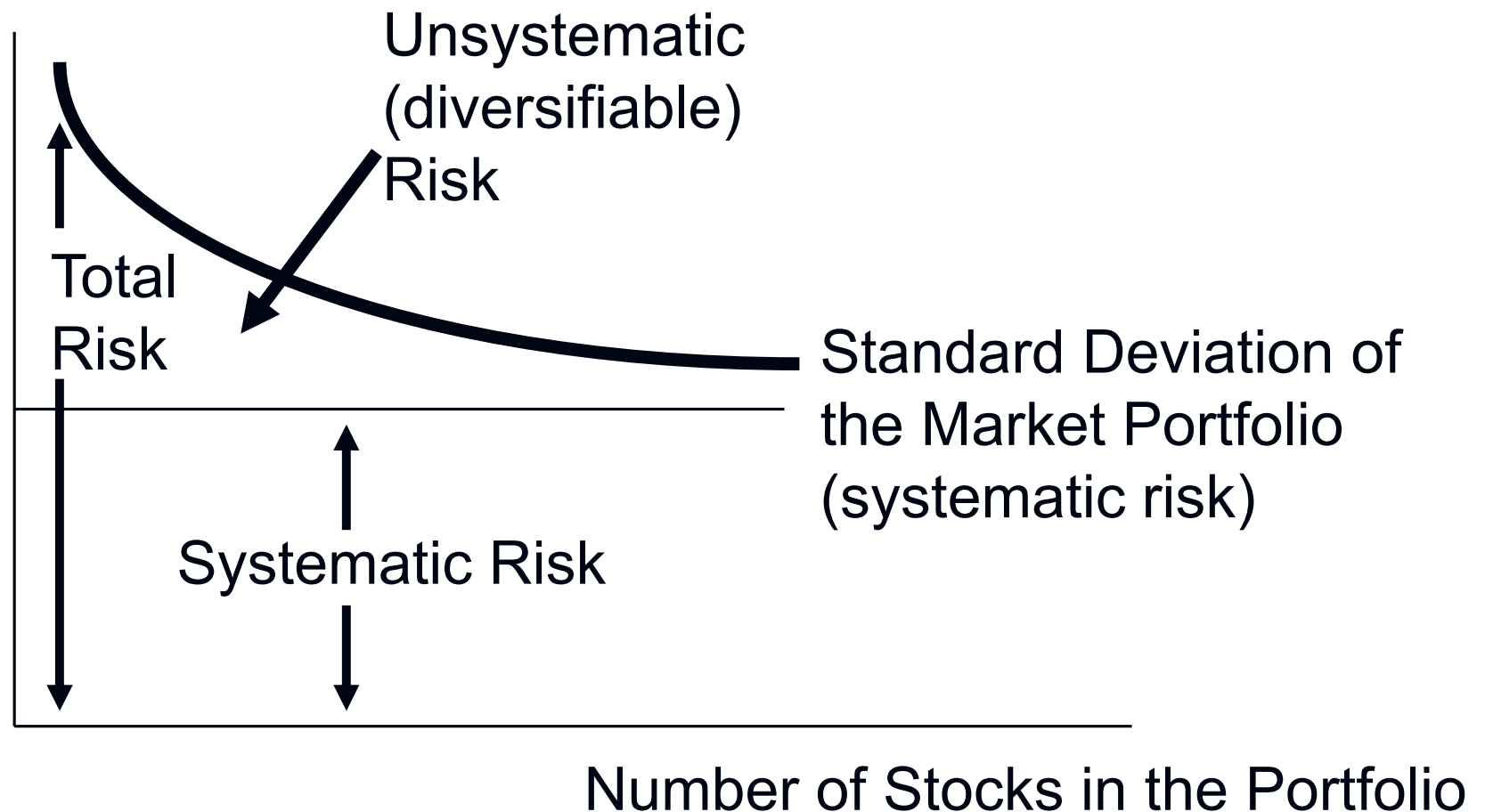


Table 13.7

(1) Number of Stocks in Portfolio	(2) Average Standard Deviation of Annual Portfolio Returns	(3) Ratio of Portfolio Standard Deviation to Standard Deviation of a Single Stock
1	49.24%	1.00
2	37.36	.76
4	29.69	.60
6	26.64	.54
8	24.98	.51
10	23.93	.49
20	21.68	.44
30	20.87	.42
40	20.46	.42
50	20.20	.41
100	19.69	.40
200	19.42	.39
300	19.34	.39
400	19.29	.39
500	19.27	.39
1,000	19.21	.39

These figures are from Table 1 in M. Statman, "How Many Stocks Make a Diversified Portfolio?" *Journal of Financial and Quantitative Analysis* 22 (September 1987), pp. 353–64. They were derived from E. J. Elton and M. J. Gruber, "Risk Reduction and Portfolio Size: An Analytic Solution," *Journal of Business* 50 (October 1977), pp. 415–37.

Total versus Systematic Risk

- Consider the following information:

	Standard Deviation	Beta
■ Security C	20%	1.25
■ Security K	30%	0.95

- Which security has more total risk? $> K$
- Which security has more systematic risk? $> C$
- Which security should have the higher expected return? $> C$ (why?)

Total Risk and Diversification

- For well-diversified portfolios, unsystematic risk is very small
- Consequently, the total risk for a diversified portfolio is essentially equivalent to the systematic risk

Example: Portfolio Beta

- Consider the previous example with the following four securities

■ Security	Weight	Beta
■ DCLK	.133	3.69
■ KO	.2	0.64
■ INTC	.267	1.64
■ KEI	.4	1.79

- What is the portfolio beta?
- $.133(3.69) + .2(.64) + .267(1.64) + .4(1.79) = 1.77$

The Capital Asset Pricing Model (CAPM)

- The capital asset pricing model defines the relationship between risk and return

$$k_s = k_{RF} + (k_M - k_{RF}) \beta$$

- If we know an asset's systematic risk, we can use the CAPM to determine its required return

Example - CAPM

- Consider the betas for each of the assets given earlier. If the risk-free rate is 4.5% and the market risk premium is 8.5%, what is the required return for each?

Security	Beta	Required Return
DCLK	3.69	$4.5 + 3.69(8.5) = 35.865\%$
KO	0.64	$4.5 + .64(8.5) = 9.940\%$
INTC	1.64	$4.5 + 1.64(8.5) = 18.440\%$
KEI	1.79	$4.5 + 1.79(8.5) = 19.715\%$